

Tyler Stahl

Forward Deployed Engineer

Glen Rock, NJ · [Portfolio](#) · [GitHub](#) · [LinkedIn](#)

SUMMARY

Forward-leaning applied AI engineer who prototypes and ships GenAI solutions with Gemini/Vertex, agentic workflows, and human-in-the-loop controls—translating ambiguous stakeholder needs into working software, from nonprofit product delivery to productionized agent systems.

EXPERIENCE

Applied AI Developer — Independent R&D

2026-02 – Present · NJ

Builds applied AI infrastructure spanning multi-agent orchestration, hybrid LLM inference, structured extraction pipelines, and human-in-the-loop controls—including the InSummery.AI family schedule concierge capstone.

- Designed a Google ADK 2.0 conditional workflow with triager and interpreter agents for registration, disruption, and schedule analysis in a Kaggle Agents Intensive capstone.
- Designed privacy-by-design masking so Gemini sees placeholders for household PII while retaining public schedule context needed for accurate extraction.
- Routed low-confidence ADK extractions through a HITL node with persisted workflow state instead of silently writing uncertain schedule data.
- Measured InSummery agent quality with deterministic scoring suites and committed model baselines instead of vibe-only demos.
- Productionized an ADK agent workflow behind Firebase Auth and Cloud Functions with observability instrumentation.
- Architected an OpenClaw multi-agent ecosystem with orchestrator and specialist agents, durable SQLite state, and structured market-data extraction.
- Built a tri-tier inference strategy separating reasoning, verification, and formatting across cloud Gemini and local Gemma models, with reported ~40% token-cost optimization.
- Developed structured-extraction pipelines that turn unstructured web data into relational formats for agent workflows.
- Added human approval gates and strict subagent controls so high-stakes agent actions require validation before commit.
- Designed a Cursor Agent SDK orchestrator that splits evaluation and testing subagents behind HITL checkpoints.
- Migrated an AI Studio personality prototype to a serverless Python backend on Firebase/Vertex AI with durable Firestore agent memory.

Technologies: Python, Google ADK 2.0, OpenClaw, Google Gemini, Gemma, Firebase, SQLite, Cursor Agent SDK

Front-End Developer — NAMI

2024-12 – 2026-01 · Remote

Built and maintained NAMI's Laravel-based core business application with a focus on reusable UI components, accessibility, and Cypress E2E testing.

- Led Cypress automated testing implementation for a Laravel/Livewire event platform, improving regression coverage and team confidence.

Technologies: Laravel, Livewire, Alpine, Cypress, JavaScript

Software Engineer — Charity Navigator

2023-08 – 2024-09 · NJ

Contributed to a high-traffic React/TypeScript data portal, GraphQL API transitions, and cross-functional delivery with product and data teams.

- Built React/TypeScript portal features for 50,000+ clients and helped sustain high-traffic events with hundreds of thousands of concurrent visitors.
- Helped transition legacy data flows into GraphQL API endpoints, streamlining backend-to-frontend communication.
- Partnered with product and data teams to ship tools that streamlined data-review processes.

Technologies: React, TypeScript, GraphQL, JavaScript

Front-End Web Developer — Charity Navigator

2019-03 – 2023-08 · NJ

Led frontend design and development for platform rebranding, technology migration, and high-traffic public web experiences.

- Partnered with the CMO and marketing team on rebrand work and time-sensitive website feature launches.

Technologies: JavaScript, HTML5, CSS, React

Founder & Web Developer — Volant Web Design

2024-09 – Present · NJ

Resumed sole-proprietorship web design and development work for B2B clients, covering requirements through production deployment and SEO.

- Delivered full-stack client websites and SEO/analytics reporting from requirements through production deployment.

Technologies: PHP, JavaScript, HTML5, CSS

SELECTED PROJECTS

InSummery.AI — Independent R&D

2026

Family scheduling information lives in messy emails, texts, and PDFs; calendar apps assume tidy manual entry and do not detect multi-child childcare gaps or real-time disruptions with a human still in control.

Built a Google ADK 2.0 multi-agent concierge that masks PII, triages and extracts schedule data, pauses below 80% confidence for HITL clarification, maintains a multi-child schedule matrix with absolute/relative gap analysis, and syncs disruptions to Google Calendar—shipped as a Kaggle + Google 5-Day AI Agents Intensive Consult-track capstone.

Technologies: Google ADK 2.0, Python, Pydantic, Google Gemini, Vertex AI, Gemma, Ollama, Firebase, Firebase Cloud Functions, Cloud Firestore, Firebase Authentication, React, Vite, Google Calendar API, OpenTelemetry

OpenClaw Multi-Agent Market Intelligence Ecosystem — Independent R&D

2025-2026

Long-running agentic market-research tasks lose state and context without durable memory and governance when synthesizing fragmented commodity and market data.

Built an autonomous research framework on OpenClaw with specialized agents (data pull, profitability analysis, dashboard UI, orchestrator), SQLite persistence, Python ETL workers, and HITL approval gates—applied to real-world and World of Warcraft commodity markets.

Technologies: OpenClaw, Python, SQLite, Google Gemini, Gemma

Cognitive Bridge: AI Personality Alignment — Independent R&D

2025-2026

Human-AI collaboration can suffer when agent personality is poorly matched to the user, and prototypes need a path from AI Studio into secured production backends.

Built an OCEAN-based personality alignment prototype in Google AI Studio, then productionized it on Firebase Cloud Functions with Firestore memory, Firebase Auth, Vertex AI, and Pydantic-validated structured Gemini outputs.

Technologies: Google AI Studio, Google Gemini, Firebase, Firebase Cloud Functions, Cloud Firestore, Firebase Authentication, Vertex AI, Pydantic, Python

Tri-Tier Hybrid Inference Pilot — Independent R&D

2025-2026

Agentic workloads need a practical balance of reasoning quality, latency, cost, and privacy across cloud and local models.

Designed a routing strategy that sends complex reasoning to stronger cloud/local models and offloads formatting and utility work to faster, cheaper models via Ollama-hosted local Gemma and cloud Gemini.

Technologies: Google Gemini, Gemma, Ollama, Antigravity, Python

AgentOS: Chief of Staff Agent — Independent R&D

2025-2026

Complex long-running workflows need reliable orchestration with mandatory human checkpoints.

Built a multi-agent orchestrator with the Cursor Agent SDK, modular subagents, and HITL handoffs for evaluation and testing stages.

Technologies: Cursor Agent SDK, Python

Charity Navigator Data Review Portal — Charity Navigator

2023-2024

Internal and client data-review workflows needed a scalable React/TypeScript portal with modern API integration.

Engineered portal features and helped transition legacy data flows into GraphQL endpoints while collaborating with product and data teams.

Technologies: React, TypeScript, GraphQL

SKILLS

Applied AI: Multi-agent orchestration, Hybrid LLM inference, Human-in-the-loop controls, PII masking, Agent evaluation, Embeddings / RAG, Firebase, Vertex AI, Ollama, Cursor Agent SDK

Front-End: JavaScript, React, TypeScript

Back-End: Python, GraphQL, SQLite

Testing: Cypress

Collaboration: Stakeholder collaboration

Generated from resume-system source data — Forward Deployed Engineer (tyler-stahl-deloitte-fde-frontier-genai)